# Lecture 02 — Vector arithmetic and geometry

**Wed Aug 26**

Last updated: August 20, 2026.

**Boyd video (optional):** Lecture 4 — addition & scalar multiplication · ENGR108 playlist (Boyd)

**Textbook:** Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 1.2–1.3. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

## Learning objectives

By the end of class, students should be able to:

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in $\mathbb{R}^2$ and recognize corresponding operations in $\mathbb{R}^3$.
4. Check whether an answer has sensible dimensions and units.

## Warm-up

Yesterday's highs (Mon–Thu), in °F, were

$$t = (63, 59, 61, 66).$$

What is $n$? What is $t_2$, and what does it mean? Is 61 a scalar or a vector? Write the 4-vector of "each day five degrees cooler than $t$" if you already know how — otherwise leave it blank and we will return to it.

## Fill in

If $a$ and $b$ are $n$-vectors, the sum $a + b$ is the $n$-vector whose $i$th entry is _____. Addition is only defined when _____.

The identity for addition is _____: $a + 0 = a$. The difference $a - b$ is _____.

If $\beta$ is a scalar, the scalar–vector product $\beta a$ is the $n$-vector whose $i$th entry is _____. In particular, $(-1)a$ is written _____.

Geometry in $\mathbb{R}^2$: draw $a$ as an arrow. Then $\beta a$ with $\beta > 0$ points _____ and has length scaled by _____. If $\beta < 0$, the arrow points _____. The sum $a + b$ can be drawn tip-to-tail or as the diagonal of a _____.

The same formulas work entrywise in $\mathbb{R}^3$; we just cannot sketch as easily on paper.

Adding a scalar to a vector is _____ in the mathematics of this course (even if some software will do it). Concatenating $(1, 2)$ with $(3, 4)$ is also _____ vector addition.

## Worked in class

**W1.** A point on the quad is $p = (2, 1)$ (east, north, in 10 m units). A walk is the displacement $a = (3, -1)$. Compute $p + a$ and sketch $p$, $a$, and $p + a$ in $\mathbb{R}^2$. Then compute $a - p$. Does $a - p$ have the same units as $p$?

**W2.** Two rain gauges, in inches, over three days:

$$r = (0.40, 0, 1.10), \qquad s = (0.15, 0.20, 0.30).$$

Compute $r + s$ and $2r$. Why is $r + (1, 1)$ not a valid vector sum here? (These are 3-vectors: same arithmetic as in $\mathbb{R}^2$, just one more entry.)

**W3.** Sanity check: can you form $(4, 1) + (0, 2, 5)$? Can you form $(18 \text{ cm}, 4 \text{ mm}) + (1, 1)$ as a meaningful measurement vector?

## Your turn

**Y1.** Compute

$$\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix} \qquad \text{and} \qquad (-2)\begin{bmatrix} 1 \\ 0.5 \end{bmatrix}.$$

**Y2.** In $\mathbb{R}^2$, sketch $u = (3, 1)$ and $\frac{1}{2}u$. In words, what did scaling by $1/2$ do? What would $-u$ look like?

**Y3.** A 2-vector of temperatures in °C is $x = (10, 14)$. A classmate types $x + 5$ in software and gets $(15, 19)$. Is that the same as a vector sum $x + v$ for some 2-vector $v$? If you need "five degrees warmer each day," what vector should you add?

## Answers (Your turn)

Y1. $(5, 2, 0)$ and $(-2, -1)$.

Y2. $\frac{1}{2}u$ is the same direction, half as long. $-u$ is the opposite direction, same length.

Y3. Software added the scalar 5 to each entry; that is not a vector–vector sum. Adding five degrees each day is the vector sum $x + (5, 5)$.